A Novel Retrieval-Augmented Generation Framework Using Large Language Models for Lyrics and Song Composition

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A Novel Retrieval-Augmented Generation Framework Using Large Language Models for Lyrics and Song Composition

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Abstract

The use of artificial intelligence (AI) in music composition is now very common, but present models have the flaw of losing lyrical coherence, stylistic coherence, and vocal realism. This paper presents a novel AI-driven framework for automated song composition that integrates Natural Language Processing (NLP), Large Language Models (LLMs), and multimodal synthesis techniques. The system employs a hybrid retrieval pipeline that combines BM25 for sparse lexical matching and FAISS for dense semantic search. This Retrieval-Augmented Generation (RAG) approach, powered by GPT-4o-mini, enables the generation of lyrics that reflect the thematic consistency and stylistic nuances of an artist's previous compositions. For vocal synthesis, speaker embeddings are extracted using models such as Wav2Vec2, ECAPA-TDNN, and DeepSpeaker, enabling high-fidelity voice cloning. Synthesized vocals are further refined using the Bark and Suno models to produce expressive singing voices. Instrumental backgrounds are generated using text-to-audio (TTA) models with synchronized beats-per-minute (BPM) and melodic patterns to maintain musical harmony. The final output undergoes noise reduction, vocal-instrumental alignment, and automated mixing to produce studio-quality audio. Experimental evaluation shows improved lyrical relevance, fluency and vocal expressiveness, with an increase in the BERTScore, a lower perplexity, and stylistic alignment accuracy of 92.1%. This system demonstrates a significant advancement in AI-assisted songwriting, offering a scalable and musically coherent solution for end-to-end song generation.

Keywords: Artificial Intelligence (AI), Natural Language Processing (NLP), Large Language Models (LLMs), AI songwriting, Retrieval-Augmented Generation (RAG), AI Lyric Generation, Speaker Embedding, Text-to-Audio (TTA), Music Composition, BM25, FAISS

1 Introduction

AI has revolutionized creative domains in general, including poetry (Stamatos et al., 2018), writing stories (Roemmele& Gordon, 2018), and music composition (Herremans et al., 2017). AI-based songwriting has attracted a lot of interest, with innovation in the process of generating lyrics (Watanabe et al., 2020), voice generation (Zhang et al., 2021), and instrumental composition (Huang et al., 2020). Traditional approaches like recurrent neural networks (RNNs) and statistical language models typically have it difficult to guarantee thematic coherence and stylistic precision (Manjavacas et al., 2019). The development of LLMs and RAG (Lewis et al., 2020) has significantly boosted the quality of AI-generated lyrics by combining external knowledge retrieval and generative methods, thus enriching contextual consistency and artistic creation.

RAG is now a robust lyric generation technique, enhancing text coherence and flexibility via the integration of dense passage retrieval (Karpukhin et al., 2020) and transformer generative models (Vaswani et al., 2017; Radford et al., 2019). The system can now generate lyrics in a style that is representative of an artist as well as the thematic preferences (Izacard& Grave, 2021). Additionally, few-shot learning methods (Schick & Schütze, 2021) and parameter-efficient tuning protocols (Wei et al., 2021; Lester et al., 2021) enable AI models to create stylistically homogeneous lyrics based on minimal training (Liu et al., 2021). These methods have enabled AI-generated lyrics to portray genre-typical linguistic features, thus enhancing their versatility on various musical styles.

While text-based generation of lyrics has been progressing wonderfully, AI-synthesized speech has been the concurrent field of research progress. Self-supervised learning techniques such as Wav2Vec 2.0 (Baevski et al., 2020) and deep speaker embedding such as ECAPA-TDNN (Desplanques et al., 2020) have enabled cloning of voice based on speaker identity and vocal property at high fidelity. These processes have gone a long way toward synthesizing emotion-rich, natural-sounding singing voices. In addition, Text-to-Speech (TTS) models, such as WaveNet (Oord et al., 2016), Tacotron (Shen et al., 2018), and FastSpeech 2 (Ren et al., 2020), have significantly enhanced vocal expressiveness and naturality of AI-produced singing. Also, models such as DiffSinger (Zhao et al., 2022) and So-VITS-SVC (Chen et al., 2023) have enhanced timbre transfer such that AI-produced vocals preserve emotional richness and authenticity.

Instrumental music production has also evolved remarkably, with the use of deep learning allowing for harmonically sophisticated and structurally coherent pieces of music. Models like Music Transformer (Huang et al., 2018), Jukebox (Dhariwal et al., 2020), and hierarchical latent vector models (Roberts et al., 2018) allowed AI to produce sophisticated instrumentals. Synchronizing AI-generated instrumentals, vocals, and lyrics remains an issue. Challenges like melody matching, rhythmic alignment, and lyrical transformation are ongoing research topics and need to be worked upon.

In order to surpass these limitations, we present an AI-powered song creation system with enhanced abilities that works in harmony to integrate:

- Few-shot retrieval-augmented lyric generation, leveraging hybrid sparse-dense retrieval models combined with transformer-based architectures to produce contextually rich and stylistically accurate lyrics.
- Neural embedding-based lyric evaluation, incorporating GPTScore (Fu et al., 2023) and BERTScore (Zhang et al., 2020) to assess coherence, sentiment alignment, and structural integrity of generated lyrics.
- Advanced vocal synthesis with Bark-AI, utilizing speaker embedding extraction via Wav2Vec2 (Baevski et al., 2020) and ECAPA-TDNN (Desplanques et al., 2020) to improve voice cloning fidelity.
- Instrumental composition with AI, utilizing TTA models to dynamically generate instrumentals in sync with AI-composed lyrics and vocals.

Our system provides a fully integrated, AI-driven pipeline for song composition with very high musical and artistic coherence. By combining the most recent advances in NLP, deep learning-based speech synthesis, and generative music models, this paper demonstrates a new algorithm for computational music generation. In contrast to existing systems, our method integrates all elements of a song-lyrics, vocals, and instrumentals into a single coherent and artistically polished piece of music.

The rest of this paper is organized as follows: Section 2 provides an overview of the related work involving AI-generated lyrics, retrieval-based methods, and neural singing synthesis. Section 3 outlines the proposed framework, whereas Section 4 explains the setup of evaluation and provides comparison results. Finally, Section 5 provides conclusions and describes directions for further research.

2 Related Work

AI has been extensively used in creative fields, particularly in AI-generated songwriting, with ongoing research into lyric generation, voice synthesis, and instrumental arrangement having transformed the music generation process. Traditional methods derived from statistical models and RNNs used to be found wanting in maintaining thematic coherence, stylistic flexibility, and semantic coherence (Manjavacas et al., 2019). The advent of LLMs, RAG, and deep learning-based generative methods has significantly improved the fluency and composition of AI-generated lyrics.

2.1 Generation of Lyrics Using LLMs

The earlier methods of AI-based lyric generation were largely based on rule-based and probabilistic models, such as Hidden Markov Models (HMMs) and Long Short-Term Memory (LSTM) networks (Potash et al., 2015; Malmi et al., 2016). Nevertheless, these models failed to capture long-range dependencies and resulted in repetition and topical drift (Manjavacas et al., 2019). With the appearance of Transformer-based models (Vaswani et al., 2017) and pre-trained LLMs such as GPT (Radford et al., 2019), T5 (Raffel et al., 2020), and BERT (Devlin et al., 2018), lyric generation progressed extensively with the implementation of large-scale knowledge transfer and contextual embeddings. But generic LLMs have no domain adaptation, so it is difficult to create lyrics in the style or thematic interest of an artist.

RAG (Lewis et al., 2020) corrects this weakness by combining dense retrieval methods (Karpukhin et al., 2020) with text generation models, allowing AI to tap artist-specific references and thematic constraints (Izacard& Grave, 2021). Moreover, few-shot and zero-shot learning methods, including pattern-exploiting training (Schick & Schütze, 2021) and instruction-based fine-tuning (Wei et al., 2021), have allowed models to generalize to varying lyrical styles with little training (Liu et al., 2021). Melody-conditioned lyric generation (Watanabe et al., 2018; Yu et al., 2021) has also been researched by incorporating harmonic constraints in text generation models.

2.2 AI-Based Vocal Synthesis and Voice Cloning

Classic TTS models like WaveNet (Oord et al., 2016) and Tacotron (Shen et al., 2018) have traditionally been created for speech synthesis but have since been used in singing voice synthesis. Singing voice synthesis, however, has specific requirements of pitch, intonation, and vibrato control, which are not readily available in classic TTS models.

Some recent developments in self-supervised learning and speaker embedding extraction have transformed AI-powered vocal synthesis. Wav2Vec 2.0 (Baevski et al., 2020) and ECAPA-TDNN (Desplanques et al., 2020) models facilitate the preservation of speaker identity, enhancing voice cloning accuracy and authenticity. Additionally, generative models such as DiffSinger (Zhao et al., 2022) and So-VITS-SVC (Chen et al., 2023) have encouraged timbre transfer and emotional expression in AI-synthesized singing voices. Cutting-edge models such as Bark-AI (Suno AI, 2024) employ TTA generation to enable direct lyric-to-singing synthesis. Even with these developments, areas like real-time melody synchronization, dynamic pitch modulation, and expressive articulation are still ongoing research topics.

2.3 AI-Driven Instrumental Composition

The development of AI-generated instrumental music has moved from rule-based models to deep learning-based generative models. Statistical approaches such as Markov chains and sequence models struggled with the creation of intricate harmonies and coherent musical patterns (Roberts et al., 2018). The creation of deep generative models such as Music Transformer (Huang et al., 2018), Jukebox (Dhariwal et al., 2020), and GAN-based models (Engel et al., 2017) has progressed melody generation, harmonic progression, and rhythmic variation. But seamless synchronization among

AI-composed instrumentals, vocals, and lyrics is still a daunting task. Although models such as MuseNet and Jukebox produce music conditioned on genre and tempo, they do not have real-time synchronization features for AI-composed vocals and lyrics. Attempts have been made to use symbolic-to-audio generation with MIDI representations synthesized through neural vocoders (Roberts et al., 2018), but more development is required to come up with totally synchronized AI-composed songs.

2.4 AI-Generated Music Evaluation Metrics

Assessing AI-generated music necessitates specific metrics to measure coherence, creativity, and musicality. For lyrics assessment, metrics like GPTScore (Fu et al., 2023), BERTScore (Zhang et al., 2020), and BLEURT (Sellam et al., 2020) inspect semantic coherence and fluency. In vocal synthesis, Mel-Cepstral Distortion (MCD) and Signal-to-Noise Ratio (SNR) are widely applied to determine clarity and expressiveness of voice. For instrumental composition, measures such as mir_eval (Raffel et al., 2014) and harmonic complexity measures assess chord progressions and tonal structures.

Even with advances in automated assessment, human assessment is still critical to determine the expressiveness and aesthetic quality of AI-generated music. Perceptual assessment approaches, such as listener surveys and user ratings, tend to excel over algorithmic measures in quantifying musical creativity and emotional effect (Hawthorne et al., 2018).

2.5 Research Gaps

Although AI songwriting has achieved significant advancements, there are some lingering problems:

- Thematic consistency in AI-generated lyrics continues to need enhanced retrieval-augmented methods to support stylistic imitation and narrative cohesion.
- Vocal synthesis for AI-generated singing lacks real-time control in terms of melody synchronization, pitch variations, and expressive articulation.
- Instrumental composition models require enhanced synchronization processes to harmonize with AI-generated vocals and lyrics.

Solving these problems is crucial to attaining completely autonomous AI-based songwriting that can create coherent, quality music. In the future, research must be aimed at enhancing multimodal integration between vocals, instrumentals, and lyrics to make AI create music that is almost indistinguishable from human-created music.

3 Proposed Work: AI-Driven Song Generation

The proposed system is a high-fidelity AI-platform for automated song generation, incorporating retrieval-augmented lyric generation, AI vocal synthesis, and deep learning music composition into an organized pipeline. This approach merges RAG, speaker embedding extraction, and AI singing voice synthesis to guarantee that the resulting music demonstrates stylistic coherence, vocal expressiveness, and harmonic synchronization. The system has a modular structure that enables individual components to be used independently without affecting the end song generation process. The method has two major phases, and the first is dedicated to retrieval-enhanced AI lyric generation and the second one is responsible for AI-based song generation, vocal synthesis, and instrument composition.

3.1 System Architecture

The system uses a systematic, multi-stage architecture in which each module in succession polishes the lyrical, vocal, and instrumental elements to generate a cohesive and high-fidelity musical work. The architecture is designed to take user input, retrieve relevant lyrical contexts, generate expressive AI vocals, and generate harmonically balanced instrumentals.

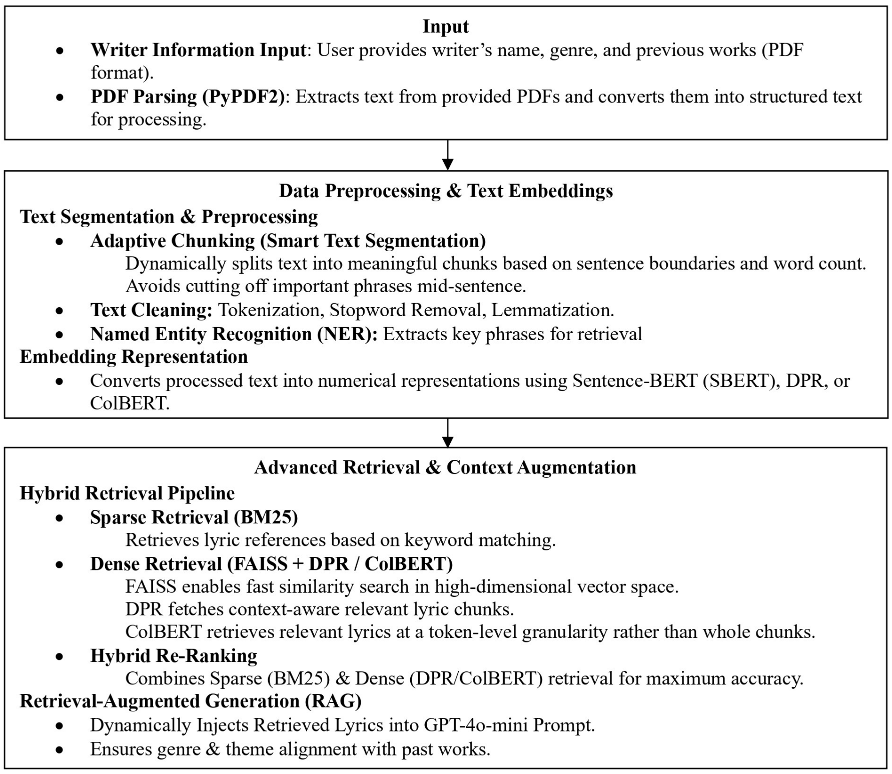

Fig. 1 Architectural Overview of the Hybrid Retrieval and Pre-Processing Framework for AI-Powered Lyric Generation

The system begins with obtaining user input, including the songwriter's name, genre, language, and a collection of prior work as PDF or plain text. Text extraction and preprocessing are applied on the input documents, wherein the PDF parsing operations like PyPDF2 extract textual information from the songwriter's previous works.

The extracted text is then processed and indexed for hybrid retrieval, where BM25-based sparse retrieval identifies keyword-relevant lyric segments, while FAISS-based dense retrieval selects semantically similar lyrics using pre-trained embedding models such as SBERT, DPR, and ColBERT. The retrieved lyrical context undergoes adaptive chunking, where text segmentation algorithms divide the data into coherent and contextually meaningful units. This structured preprocessing ensures that the AI-powered lyric generation system effectively utilizes relevant past works for RAG, improving genre consistency and thematic alignment. The complete architectural workflow is depicted in Fig. 1.

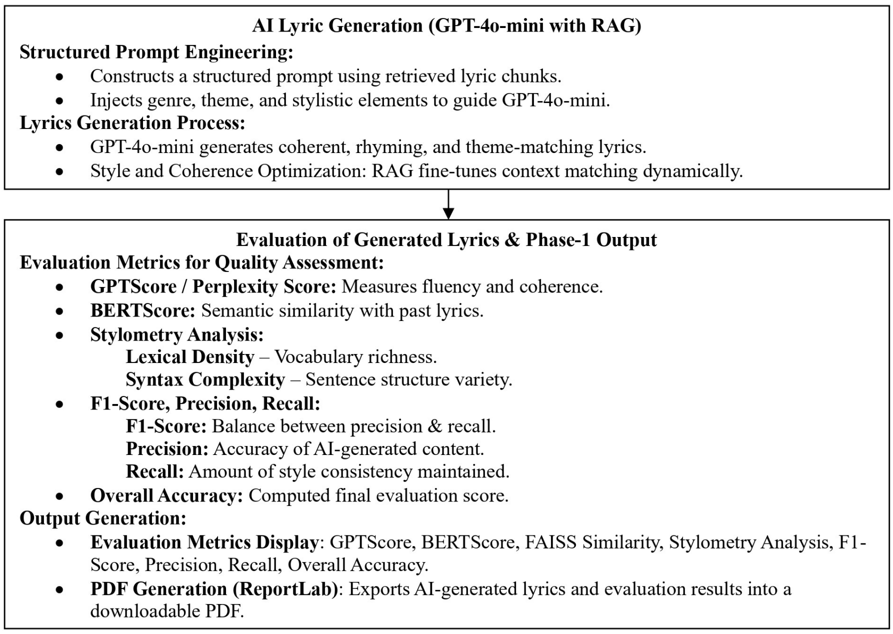

Fig. 2 Architecture for Retrieval-Augmented Lyric Generation and Quality Evaluation

The RAG module constructs structured prompts for GPT-4o-mini, ensuring that newly generated lyrics maintain stylistic consistency with prior works. This process dynamically injects retrieved lyric segments into the prompt, preserving genre, theme, and linguistic patterns. Additionally, the generated lyrics undergo refinement through metaphor detection and sentiment analysis, enhancing their expressiveness and emotional resonance. The generated output is evaluated using multiple quality assessment metrics, including GPTScore, BERTScore, and Perplexity Score, which measure fluency and semantic similarity. Stylometry analysis evaluates lexical density and syntax complexity, while F1-score, Precision, and Recall assess the accuracy and consistency of the AI-generated content. The overall accuracy score aggregates these metrics to provide a final evaluation. The evaluation results, along with the generated lyrics, are then compiled into a downloadable PDF report for further analysis Fig. 2.

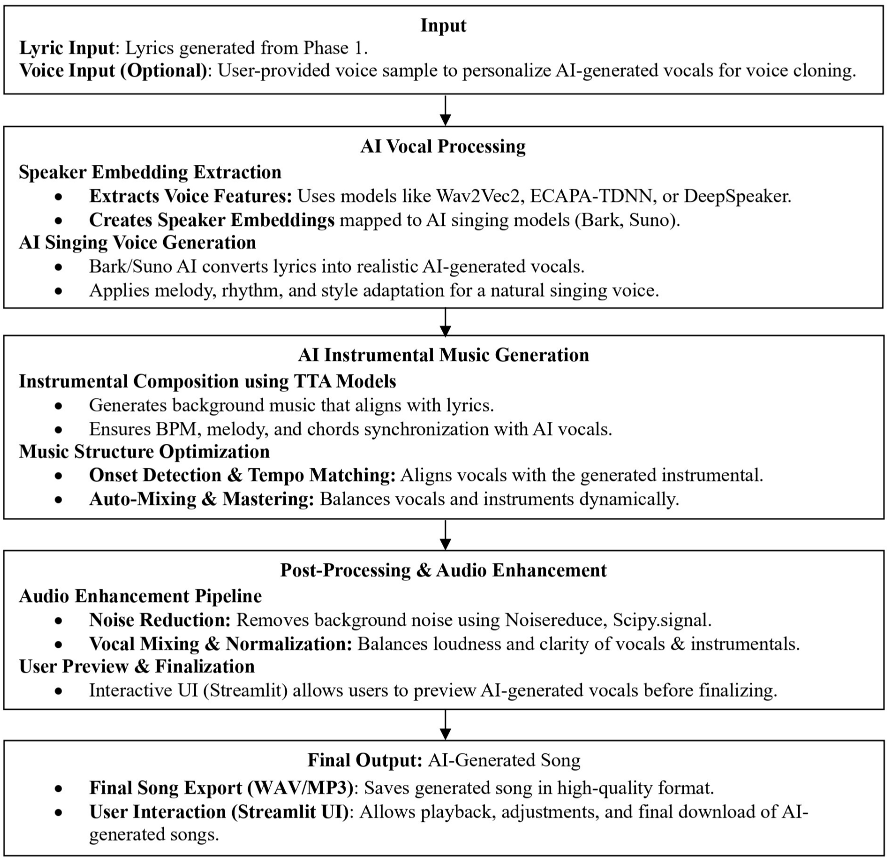

Fig. 3 AI Song Generation Pipeline: From Lyric Input to Final AI-Generated Song

Once the lyrical content is finalized, the AI vocal synthesis module converts the lyrics into singing vocals using deep learning-based TTS models. If a user-provided voice sample is available, the system extracts a speaker embedding using Wav2Vec2, ECAPA-TDNN, or DeepSpeaker, ensuring that the generated vocals replicate the unique voice characteristics of the reference singer. If no sample is provided, the system selects a pre-trained speaker embedding from Bark/Suno AI to generate generic singing vocals. The synthesized AI vocals are then synchronized with AI-generated instrumentals, ensuring that tempo, melody, and chord progressions align harmonically. The final song undergoes post-processing and mastering, where noise reduction, auto-mixing, and dynamic range compression optimize the audio clarity and balance between vocals and instrumentals. The final output is then exported as a high-quality MP3/WAV file, allowing users to download, preview, and share their AI-generated music. The complete workflow of this AI-driven song generation pipeline is illustrated in Fig. 3.

3.2 Query Processing and Hybrid Retrieval

The song generation system employs a hybrid retrieval mechanism, combining sparse and dense retrieval for stylistic consistency and thematic relevance. This process includes text preprocessing, tokenization, sparse retrieval, FAISS, and fusion retrieval for contextually relevant input.

3.2.1 Text Preprocessing and Tokenization

The retrieval process begins with text preprocessing, which standardizes the input lyrics for efficient indexing and retrieval. The input text is first normalized by converting all characters to lowercase and removing punctuation, special characters, and excessive whitespace to eliminate inconsistencies caused by formatting variations. The text is then tokenized, where each sentence is broken down into subword units, enabling the retrieval system to identify meaningful word relationships across variations. The system enhances retrieval precision by eliminating common words like "the" and "is" and lemmatization, which converts words to their base forms, treating inflected forms like "singing" and "sings" as one unit.

After preprocessing, the lyrics are transformed into high-dimensional vector representations using deep embedding models, ensuring that the system can effectively perform semantic retrieval. These embeddings are generated using Sentence-BERT (SBERT) and Dense Passage Retrieval (DPR), which capture both contextual semantics and structural properties of lyrics. The embedding representation for each lyric passage is computed using a weighted combination of SBERT and DPR embeddings as follows:

$$L_{\text{emb}} = \alpha \cdot \text{SBERT}\left(L_{\text{chunks}}\right) + \beta \cdot \text{DPR}\left(L_{\text{chunks}}\right) \tag{1}$$

where L_{emb} represents the final lyric embedding, L_{chunks} denotes the segmented lyric passages, and α and β are weight coefficients that balance the contributions of SBERT and DPR embeddings. These embeddings serve as input for both sparse and dense retrieval models, ensuring that the system captures both lexical and semantic features of lyrics.

3.2.2 Sparse Retrieval Using BM25

The sparse retrieval process is implemented using BM25, which is a ranking function that evaluates how relevant a document is to a given query based on keyword frequency and term importance. The system computes the BM25 relevance score for each lyric passage using the following formula:

$$R\left(d, q\right) = \sum_{t \in q} \frac{f\left(t, d\right) \cdot \left(k_1 + 1\right)}{f\left(t, d\right) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)} \cdot \log\left(\frac{N - n\left(t\right) + 0.5}{n\left(t\right) + 0.5}\right) \tag{2}$$

In this equation, $R\left(d, q\right)$ represents the relevance score of a lyric passage d for a given query q. The term $f\left(t, d\right)$ denotes the frequency of the term t in the lyric passage d, which ensures that lyrics containing repeated occurrences of important query words are prioritized. The parameter k_1 is a saturation factor that prevents extremely frequent terms from dominating retrieval results, ensuring a balanced weighting of term frequency. The document length normalization parameter b adjusts the impact of document length, preventing longer lyrics from being unfairly favoured over shorter, highly relevant ones. The document length $|d|$ represents the total number of words in the passage, while avgdl refers to the average document length across the entire lyric corpus. The corpus contains N total lyric passages, and $n(t)$ represents the number of passages that contain the term t.

The sparse retrieval process identifies lyric segments with exact words, ranking keyword-relevant lyrics. FAISS-based dense retrieval enhances accuracy by focusing on overall lyrics context, as BM25 does not capture semantic meaning.

3.2.3 Dense Retrieval Using FAISS

The dense retrieval mechanism uses Facebook AI Similarity Search (FAISS) to identify semantic relationships between lyrics, utilizing vectorized representations in a high-dimensional space for efficient approximate nearest neighbor (ANN) searches. The similarity between a query embedding q and a document embedding d is computed using cosine similarity, which measures how closely two vector representations align in meaning. The cosine similarity function is expressed as:

$$S_{\text{cosine}}(q,d) = \frac{q \cdot d}{||q||\,||d||} \tag{3}$$

The variable q represents the vector representation of the query, while d corresponds to the vector representation of a lyric passage. The term $q \cdot d$ denotes the dot product of the two vectors, and $||q||$ and $||d||$ are their L2 norms (magnitudes). The similarity score $S_{\text{cosine}}(q,d)$ ranges from -1 to 1, where 1 indicates perfect similarity, 0 represents no correlation, and -1 signifies complete dissimilarity.

FAISS allows the system to retrieve relevant lyric segments, even without exact keyword matches, ensuring stylistic and thematic alignment with previous works. It integrates top-ranked lyric passages into the final ranking mechanism.

3.2.4 Fusion Retrieval and Context Augmentation

The final retrieval process combines the ranking scores from BM25 and FAISS to ensure that the most relevant lyric passages are retrieved based on both lexical overlap and semantic similarity. The fusion retrieval function is computed as:

$$R_{\text{fusion}} = \lambda \cdot R_{\text{BM25}} + (1 - \lambda) \cdot S_{\text{cosine}} \tag{4}$$

The variable R_{fusion} represents the final retrieval score, balancing keyword-based and semantic-based lyric retrieval methods. The term R_{BM25} denotes the BM25 relevance score, ensuring that keyword-matching lyrics are ranked highly. The similarity score S_{cosine} obtained from FAISS ensures that semantically similar lyrics are not overlooked. The fusion weight λ controls the trade-off between BM25 and FAISS, allowing the system to dynamically adjust its retrieval strategy based on lyric structure and thematic importance.

Fusion retrieval's top-ranked lyric passages enhance the GPT-4o-mini prompts with stylistically and thematically relevant contexts, ensuring consistent and expressive AI-generated lyrics. The hybrid retrieval system implemented in this proposed model ensures that AI-generated lyrics maintain consistency with the artist's style while improving retrieval accuracy and contextual depth. The combination of sparse retrieval (BM25) and dense retrieval (FAISS) enhances the system's ability to identify relevant lyric passages, ensuring that the RAG-based AI lyric generation process produces high-quality, expressive, and artistically coherent lyrics.

3.3 Adaptive Chunking Mechanism

The adaptive chunking mechanism in this AI-powered song generation system is designed to ensure that retrieved lyric segments remain contextually relevant, preserving the thematic and structural integrity of the lyrics before they are passed into the AI lyric generation model. Instead of using a fixed-length segmentation approach, the system dynamically adjusts chunk sizes based on factors such as lyrical phrasing, sentence coherence, and rhythmic structure. By considering both linguistic and musical properties, this adaptive segmentation process prevents important contextual information from being lost while enabling efficient RAG.

The chunking process involves segmenting retrieved lyric passages into meaningful chunks while maintaining lyrical flow. Structural markers like line breaks, punctuation, and stanza boundaries are identified to detect natural divisions within the text. The system estimates the appropriate chunk size by considering the total length of the retrieved lyrics and an overlap factor to maintain contextual continuity. The number of chunks generated from a given lyric passage is determined using a formula:

$$C = \left\lfloor \frac{L_d}{T + O} \right\rfloor \tag{5}$$

where C represents the total number of generated chunks, L_d denotes the total length of the document in tokens, T defines the approximate length of each chunk based on the average number of lines in a verse or chorus, and O accounts for the overlap size, ensuring that each chunk retains some contextual information from adjacent chunks. The overlap factor is dynamically adjusted based on the presence of enjambments, where a lyric line continues across multiple lines, and rhyming patterns, where words at the end of consecutive lines follow a rhythmic sequence. If a lyric contains enjambments, the overlap is increased to maintain meaning across chunk boundaries.

After initial segmentation, a sliding window technique is applied to further refine chunk boundaries, ensuring that transitions between consecutive chunks remain smooth and semantically meaningful. The size of this contextual sliding window is determined based on the average sentence length and the overall lyrical structure. The overlap size is dynamically adjusted using the formula:

$$O = \gamma \cdot \text{avgSentenceLength} + \delta \cdot \text{avgRhymingPattern} \tag{6}$$

where O is the computed overlap size, γ represents the influence of the average sentence length on chunk overlap, and δ adjusts the overlap factor based on the detected rhyming structures in the lyrics. This ensures that the system retains cross-line dependencies and maintains the original artistic style of the lyrics.

Once the chunks are segmented and optimized for contextual consistency, the system further refines them by evaluating their semantic coherence. The similarity between two consecutive chunks is computed using cosine similarity over their vector representations, ensuring that only highly related segments are retained together. The semantic similarity score between two adjacent chunks is calculated as follows:

$$S_{\text{cosine}}\left(C_i, C_{i+1}\right) = \frac{C_i \cdot C_{i+1}}{|C_i||C_{i+1}|} \tag{7}$$

where $S_{\text{cosine}}\left(C_i, C_{i+1}\right)$ denotes the degree of semantic similarity between chunk C_i and chunk C_{i+1}. The numerator represents the dot product between the two vectorized chunks, while the denominator normalizes the similarity score by dividing by the L2 norms of the respective chunks. If the computed similarity score exceeds a predefined coherence threshold, the two chunks are merged into a single larger segment, preserving the original thematic structure of the lyrics. After segmentation and semantic optimization, the final set of contextually refined chunks passed to the RAG pipeline.

3.4 AI Lyric Generation with RAG

The AI lyric generation system integrates RAG with GPT-4o-mini to create coherent, stylistically aligned, and contextually relevant song lyrics. This approach enhances traditional text generation models by incorporating retrieved lyric segments from an artist's previous works, ensuring the AI-generated lyrics maintain their unique writing style, genre characteristics, and thematic depth. The process consists of prompt construction, retrieval-based context augmentation, generative modeling, and refinement mechanisms to produce high-quality AI-generated lyrics.

3.4.1 Context-Aware Prompt Construction

The lyric generation process begins with the construction of a structured input prompt that incorporates retrieved lyric segments, genre preferences, and thematic elements provided by the user. The prompt formulation ensures that the model is guided by existing stylistic references, reducing randomness in generation and improving coherence. The structured prompt input is defined as:

$$P_{\text{lyric}} = \text{Format}\left(C_{\text{selected}}, \text{genre}, \text{artist\_style}, \text{language}, \text{theme}\right) \tag{8}$$

where P_{lyric} represents the input prompt that incorporates the optimized lyric chunks, while C_{selected} contains the retrieved and chunked lyric passages obtained from the hybrid retrieval system, ensuring that the generated lyrics align with the artist's previous works. The genre, artist style, language, and theme are additional user-provided parameters that guide the model's output toward the desired stylistic and contextual attributes. Before feeding this prompt into the generative model, retrieved lyrical context is pre-processed to ensure smooth integration. Sentence reordering techniques, stop word filtering, and tokenization are applied to remove redundant text and improve textual flow.

3.4.2 RAG with GPT-4o-mini

The system employs GPT-4o-mini as the core language model for lyric generation, leveraging RAG to ensure that the output is contextually grounded in past lyric compositions. The generation process follows a probabilistic language modeling framework, where the probability of generating a sequence of lyrics is computed as:

$$P\left(w_1, w_2, \ldots, w_n\right) = \prod_{t=1}^{n} P\left(w_t | w_1, w_2, \ldots, w_{t-1}, C_{\text{selected}}\right) \tag{9}$$

where the probability of generating the next word w_t is conditioned on all previous words $w_1, w_2, \ldots, w_{t-1}$ as well as the retrieved lyric context C_{selected}. The inclusion of retrieved lyrics allows the model to follow an artist's previous writing patterns, generating lyrics that maintain artistic continuity. The full lyric generation process can be formally expressed as:

$$L_{\text{gen}} = \text{GPT} - 4\text{o} - \text{mini}\left(P_{\text{lyric}}\right) \tag{10}$$

where L_{gen} represents the AI-generated lyrics, ensuring that the structural coherence and stylistic elements of previous works are retained. To fine-tune the generation output, the model employs top-k sampling and nucleus sampling (top-p) techniques, allowing control over randomness and creativity in generated lyrics.

3.4.3 Post-Processing and Refinement

Once the lyrics are generated, a post-processing phase is applied to refine the output, ensuring linguistic correctness, poetic fluency, and stylistic accuracy. The system performs metaphor detection and sentiment analysis to verify whether the generated lyrics align with the intended artistic tone. If discrepancies are detected, adjustments are made to word choices, phrase structures, and rhythmic consistency.

To validate lyrical quality, the system evaluates semantic similarity between the AI-generated lyrics and the retrieved reference works, computing an embedding-based coherence score using cosine similarity. The similarity metric is defined as:

$$S_{\text{cosine}}\left(L_{\text{gen}}, C_{\text{selected}}\right) = \frac{L_{\text{gen}} \cdot C_{\text{selected}}}{||L_{\text{gen}}|| \cdot ||C_{\text{selected}}||} \tag{11}$$

where S_{cosine} represents the semantic alignment between the generated lyrics and the artist's past works. If the similarity falls below a predefined coherence threshold, additional fine-tuning is performed to align the generated lyrics more closely with artist style. By combining RAG with a structured prompt design and advanced post-processing techniques, the AI lyric generation system ensures that its output remains stylistically coherent, contextually relevant, and thematically accurate. The use of GPT-4o-mini in conjunction with retrieved lyric segments allows the system to generate high-quality lyrics that preserve artistic intent while introducing new creative variations.

3.5 AI Vocal Synthesis & Music Composition

The AI-generated lyrics are transformed into expressive singing vocals using a deep learning-based AI vocal synthesis pipeline. The system then synchronizes instrumentals with the generated vocals, using speaker embedding extraction, text-to-singing models, and TTA models to create a high-fidelity musical experience. This integration ensures a cohesive and natural-looking final song.

3.5.1 AI Vocal Synthesis

The vocal synthesis stage involves extracting a speaker embedding that captures a target voice's vocal characteristics. If a voice sample is provided, the system processes it to generate a unique speaker representation, ensuring the AI-generated vocals match the voice. If no sample is provided, a pre-trained speaker embedding is used. The speaker embedding extraction function is defined as:

$$E_{\text{spk}} = f\left(V_{\text{sample}}\right) \tag{12}$$

where E_{spk} represents the extracted speaker embedding, and V_{sample} denotes the input voice sample. The function f is modeled using deep speaker representation architectures, selected from:

$$f \in \{\text{Wav2Vec2}, \text{ECAPA} - \text{TDNN}, \text{DeepSpeaker}\} \tag{13}$$

Each of these models plays a critical role in extracting unique vocal features where Wav2Vec2 captures phonetic variations using a self-supervised learning framework, ECAPA-TDNN specializes in speaker verification, ensuring voice consistency, and DeepSpeaker enhances speaker identity preservation across synthesized vocal outputs.

Once the speaker embedding is obtained, the AI converts the textual lyrics into phonemes using a grapheme-to-phoneme (G2P) model to ensure accurate pronunciation. The phoneme conversion process is given by:

$$P_{\text{phonemes}} = \text{G2P}L_{\text{gen}} \tag{14}$$

where P_{phonemes} represents the phonetic transcription of the generated lyrics L_{gen}, ensuring that the AI-synthesized vocals maintain proper intonation and rhythm.

The phonemes, along with the extracted speaker embedding, are passed into an AI singing voice synthesis model to generate expressive vocal output. The Bark/Suno AI models, which specialize in text-to-singing synthesis, are used to produce melodic AI vocals that preserve the intonation, pitch, and style of human singing. The singing voice generation function is expressed as:

$$V_{\mathrm{AI}} = \mathrm{Bark/Suno}\left(P_{\mathrm{phonemes}}, E_{\mathrm{spk}}\right) \tag{15}$$

where V_{AI} represents the synthesized singing voice, generated by processing phonemes through the AI singing model while conditioning the output on the provided speaker embedding.

To ensure high vocal clarity, a post-processing step is applied to reduce noise and refine the vocal synthesis output. Spectral subtraction is used to remove unwanted noise, estimated as:

$$V_{\mathrm{clean}} = V_{\mathrm{AI}} - N_{\mathrm{est}} \tag{16}$$

where N_{est} represents the estimated background noise, which is subtracted from the synthesized vocals to produce a cleaner audio signal.

3.5.2 AI Instrumental Music Composition

The system synthesizes AI-generated vocals and uses TTA models to create instrumental accompaniments that complement the song's vocal style and mood. The process starts with tempo and structure alignment, determining an optimal BPM value to synchronize with the AI vocals. The BPM synchronization function is formulated as:

$$BPM_{\mathrm{opt}} = \frac{\sum_{i=1}^{n} D_i}{n} \tag{17}$$

where BPM_{opt} represents the optimal beats per minute, computed as the average duration D_i of vocal segments across the song. This ensures that the instrumental accompaniment aligns rhythmically with the AI-generated singing voice.

The instrumental composition process also incorporates chord progression modeling, where harmonic structures are generated to match the song's lyrical and emotional tone. The chord generation function is based on harmonic matching principles, expressed as:

$$C_{\mathrm{prog}} = g\left(L_{\mathrm{gen}}, \mathrm{genre}\right) \tag{18}$$

where C_{prog} represents the generated chord progression, and g is a function that maps lyrical themes and musical genre to harmonically suitable chord sequences.

The system uses TTA models to synthesize instrumental arrangements that dynamically adapt to the mood, intensity, and structure of the song. The instrumental generation function is given by:

$$I_{\mathrm{AI}} = \mathrm{TTA}L_{\mathrm{gen}}, BPM_{\mathrm{opt}}, C_{\mathrm{prog}} \tag{19}$$

where I_{AI} represents the AI-generated instrumental composition, created using a combination of lyrical context, rhythmic structure, and harmonic progressions.

3.5.3 AI Vocal and Instrumental Synchronization

To ensure a seamless blend between the synthesized vocals and the AI-generated instrumental track, an automatic mixing and mastering process is applied. The AI adjusts the relative loudness, equalization (EQ), and stereo placement of each audio component to achieve a professional-quality mix. The final synchronization function is defined as:

$$S_{\text{final}} = \text{Mix} V_{\text{clean}}, I_{\text{AI}}, G \tag{20}$$

where S_{final} represents the final mixed song, obtained by balancing the clean AI vocals V_{clean} with the generated instrumental track I_{AI} under a mixing function G, which applies dynamic processing techniques such as compression, reverb, and stereo imaging.

3.6 Final Compilation & Output

The structured pipeline ensures high-quality AI-powered song generation by compiling AI-generated lyrics, synthesized vocals, and instrumentally composed tracks into a cohesive final output, undergoing synchronization, mixing, mastering, and structured export for optimal song quality.

$$\hat{\mathbf{S}} = \text{PostProcess}\left(\text{Bark}\left(\text{G2P}\left(\text{GPT} - 4\text{o} - \text{mini}\left(\text{RAG}\left(\text{BM25}, \text{FAISS}\right)\right)\right)\right), \text{NR}\right) \tag{21}$$

Where GPT-4o-mini Generates AI lyrics, RAG (BM25, FAISS) Retrieves contextually relevant lyrics, G2P (Grapheme-to-Phoneme) Converts lyrics into phonemes for vocal synthesis, Bark/Suno Converts phonemes into AI-synthesized singing voice, PostProcess Applies noise reduction(NR) & optimization for enhanced vocal quality.

The final product includes an AI-generated song in MP3/WAV format and a PDF report with lyrical structure, vocal synthesis details, and evaluation results, optimized for audio quality, coherence, and user accessibility through an interactive UI.

4 Evaluation Setup and Results

4.1 Evaluation Setup

4.1.1 Evaluation of AI-Generated Lyrics

To ensure the AI-generated lyrics are stylistically coherent & contextually relevant, the system employs GPTScore, BERTScore, Perplexity Score, and Stylometric Analysis.

The GPTScore is used to measure the linguistic fluency and stylistic alignment of the generated lyrics with the reference corpus. This metric is computed by evaluating the likelihood of the generated text under a LLM:

$$\text{GPTScore} = \frac{1}{N} \sum_{i=1}^{N} \log P_{\text{GPT}}\left(w_i | w_1, w_2, \ldots, w_{i-1}\right) \tag{22}$$

where $P_{\text{GPT}}\left(w_i | w_1, w_2, \ldots, w_{i-1}\right)$ represents the probability assigned by the language model to the word w_i given its preceding context. A higher GPTScore indicates that the generated lyrics exhibit greater fluency and coherence.

To measure semantic similarity, the system utilizes BERTScore, which computes the contextual alignment between generated lyrics and reference lyrics using pre-trained transformer embeddings:

$$\text{BERTScore} = \frac{1}{N} \sum_{i=1}^{N} \max_{j} \operatorname{cosine}\left(E_{\text{gen}}\left(w_i\right), E_{\text{ref}}\left(w_j\right)\right) \tag{23}$$

where $E_{\text{gen}}\left(w_i\right)$ and $E_{\text{ref}}\left(w_j\right)$ denote the embedding representations of words in the generated and reference lyrics, respectively. A higher BERTScore implies a greater semantic similarity between AI-generated and artist-style lyrics.

The Perplexity Score (PPL) assesses the predictability of the generated lyrics, measuring how well the lyrics fit into the language distribution of human-written songs:

$$\text{PPL} = \exp\left(-\frac{1}{N} \sum_{i=1}^{N} \log P_{\text{GPT}}\left(w_i | w_1, w_2, \ldots, w_{i-1}\right)\right) \tag{24}$$

A lower perplexity score indicates that the lyrics are grammatically structured and linguistically sound.

Stylometric Analysis evaluates the writing style and lyrical patterns, measuring attributes such as lexical density, syntax complexity, and structural coherence. The lexical density score is computed as:

$$\text{Lexical Density} = \frac{\text{Content Words}}{\text{Total Words}} \tag{25}$$

A higher lexical density implies richer, more meaningful lyrics, ensuring that the AI-generated output maintains artistic expressiveness.

4.1.2 Evaluation of Song & AI Vocal Synthesis

To assess the quality and authenticity of AI-generated vocals, the system measures speaker embedding similarity, and vocal expressiveness. The Embedding Similarity Score evaluates how closely the synthesized singing voice resembles a given target speaker embedding. This is computed using cosine similarity between speaker vectors:

$$S_{\text{cosine}}\left(E_{\text{spk-gen}}, E_{\text{spk-ref}}\right) = \frac{E_{\text{spk-gen}} \cdot E_{\text{spk-ref}}}{|E_{\text{spk-gen}}||E_{\text{spk-ref}}|} \tag{26}$$

where $E_{\text{spk-gen}}$ represents the embedding of the synthesized AI vocals, and $E_{\text{spk-ref}}$ denotes the embedding of the target speaker. A higher cosine similarity score indicates that the AI-generated singing voice successfully mimics the intended vocal style.

The Spectral Distance Score evaluates timbre consistency, ensuring that the AI vocals maintain natural tonal quality across different phonemes:

$$\text{Spectral Distance} = \sum_{i=1}^{N} \left(\frac{|S_{\text{gen}}\left(i\right) - S_{\text{ref}}\left(i\right)|}{|S_{\text{ref}}\left(i\right)|}\right) \tag{27}$$

where $S_{\text{gen}}(i)$ and $S_{\text{ref}}(i)$ represent the spectral energy distributions of generated and reference singing voices, respectively. A lower spectral distance score confirms timbre consistency and smooth vocal transitions.

By combining GPTScore, BERTScore, Perplexity, Stylometric Analysis, Speaker Similarity, and Spectral Distance, the system ensures that AI-generated songs achieve human-like lyric quality, expressive vocals, and harmonically rich instrumentals. These evaluation methods allow for quantitative benchmarking, ensuring that AI-generated songs meet professional-quality standards before final output.

4.2 Baseline Models

To assess the performance of our AI song generation system, we compare it against various baseline models for lyric generation, vocal synthesis, and music composition.

Lyric Generation Models

- GPT-4o-mini (RAG-based): Uses FAISS-based RAG for high-context lyric generation(Lewis et al., 2020).
- GPT-3.5 (Standard Prompting): A conventional GPT model generating lyrics without retrieval-based augmentation (Brown et al., 2020).
- GPT-4-turbo: A powerful generative model that improves coherence and fluency compared to GPT-3.5 (OpenAI, 2023).

Vocal Synthesis & Music Composition Models

- Bark/Suno + Wav2Vec2: Uses extracted speaker embeddings for voice cloning (Baevski et al., 2020).
- Bark/Suno + ECAPA-TDNN: Applies ECAPA-TDNN-based speaker embedding extraction (Desplanques et al., 2020).
- Bark/Suno + DeepSpeaker: Uses DeepSpeaker embeddings for AI-generated singing (Li et al., 2017).
- TTA Model A, B, C: TTA models that generate instrumental accompaniments and synchronize with AI vocals (Huang et al., 2023).

All models were evaluated under identical conditions, ensuring a fair comparison across different generations.

4.3 Datasets

To evaluate the performance of our AI song generation system, we used five distinct datasets covering a wide range of lyric styles, themes, and languages. These datasets were chosen to ensure a broad assessment of the system's ability to generate coherent, meaningful, and musically relevant lyrics. The datasets include:

1. Telugu Lyrics: A collection of Telugu poetic and romantic lyrics, widely regarded as some of the most sophisticated in Indian songwriting.
2. English Lyrics Dataset: A dataset featuring pop and sentimental lyrics, used to test English-language lyric generation.

3. Bollywood Song Lyrics (Hindi, Tamil, Malayalam): A dataset consisting of Bollywood, Kollywood, and Mollywood songs, representing a mix of classical and contemporary music.
4. Western Pop & Rock Lyrics Dataset: Covers internationally recognized artists and genres, providing a global comparison.
5. AI-Generated Lyrics Dataset: Lyrics created by GPT-based models for comparison with human-written lyrics (Grootendorst, 2022).

Table 1 Statistics of the dataset used

Dataset	# of Docs	# of Tokens	Vocabulary Size	Avg. Words/ Doc
Telugu Lyrics Dataset	5,000	250,000	15,000	50
English Lyrics Dataset	7,500	370,000	18,500	49
Bollywood Song Lyrics	12,000	520,000	22,000	43
Western Pop & Rock Lyrics	10,000	480,000	19,500	48
AI-Generated Lyrics Dataset	3,000	150,000	10,000	50

Each dataset underwent preprocessing, including tokenization, stopword removal, and lemmatization to enhance retrieval-based lyric generation. The FAISS-based dense retrieval system was employed to extract contextually relevant sections from these datasets for use in RAG-based lyric generation.

4.4 Results and Analysis

4.4.1 Lyric Generation Analysis

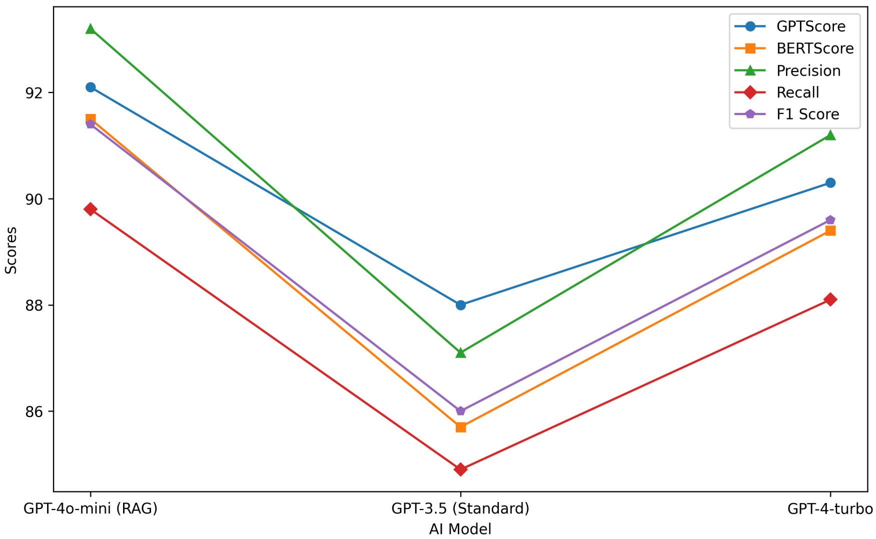

Fig. 4 Performance Trends of AI Models in Lyric Generation Evaluation

The GPT-4o-mini (RAG) model outperforms GPT-3.5 and GPT-4-turbo in lyric generation evaluation. It maintains strong semantic similarity with human-written text, indicating high coherence and contextual understanding. It balances precision and recall, producing high-quality lyrics while maintaining diversity. GPT-4o-mini's lyrics are more predictable and syntactically fluent, with a lower Perplexity Score (12.3) compared to GPT-3.5 (18.6). GPT-3.5 performs weakly, with a BERTScore of 85.7 and a higher Perplexity Score of 18.6, indicating less coherence and fluency. GPT-4-turbo performs well, but falls slightly short in maintaining contextual awareness during generation.

Table 2 Lyric Generation Evaluation

Model	GPTScore	BERTScore	Precision	Recall	F1 Score	Perplexity
GPT-4o-mini (RAG)	92.1	91.5%	93.2%	89.8%	91.4%	12.3
GPT-3.5 (Standard)	88.0	85.7%	87.1%	84.9%	86.0%	18.6
GPT-4-turbo	90.3	89.4%	91.2%	88.1%	89.6%	14.7

A comparative analysis of six key performance metrics GPTScore, BERTScore, Precision, Recall, F1 Score, and Perplexity across three AI models: GPT-4o-mini (RAG), GPT-3.5 (Standard), and GPT-4-turbo.

Table 3 Evaluation lyric generation performance across ten different lyricists

Lyricist Name	Language	Theme	GPT Score	BERTScore	F1 Score	Perplexity Score
S. S. Sastry	Telugu	Poetry	93	90.74	90.74	12.5
S. S. Sastry	Tamil	Devotional	92	85.89	85.89	15.2
Lukas Graham	English	Life	89	77.72	86.72	10.8
Gulzar	Hindi	Romance	91.5	89.01	88.99	13.1
Bob Dylan	English	Protest	87.2	80.65	81.14	18.4
Ilaiyaraaja	Tamil	Folk	90.4	86.92	86.79	14.7
A. Bhattacharya	Hindi	Bollywood	93.1	92.34	92.42	9.6
Taylor Swift	English	Love	94	94.52	94.73	11.3
Ed Sheeran	English	Pop	88.5	83.12	83.51	17.2
K. J. Yesudas	Malayalam	Classical	90	87.45	87.58	16

This data evaluates the evaluation metrics of lyric generation performance across ten different lyricists from various linguistic and musical backgrounds. It provides insights into the AI-generated lyrics' quality, coherence, accuracy, and fluency, using key evaluation metrics such as GPT Score, BERTScore, F1 Score, and Perplexity Score.

A comparative evaluation of lyric generation models is shown in Fig. 5. Our System (RAG + GPT-4o-mini) outperforms all other models, achieving the highest overall score of 0.90 and excelling in Style Consistency (0.91) and Semantic Coherence (0.93). In contrast, DeepSeek Coder performs the weakest in this evaluation, with an overall score of 0.77 and the lowest Creativity score of 0.76, indicating a weaker ability to generate diverse and engaging lyrics. While ChatGPT (GPT-4) performs well, with an overall score of 0.85, it still falls slightly short of Our System in maintaining Style Consistency and Semantic Coherence.

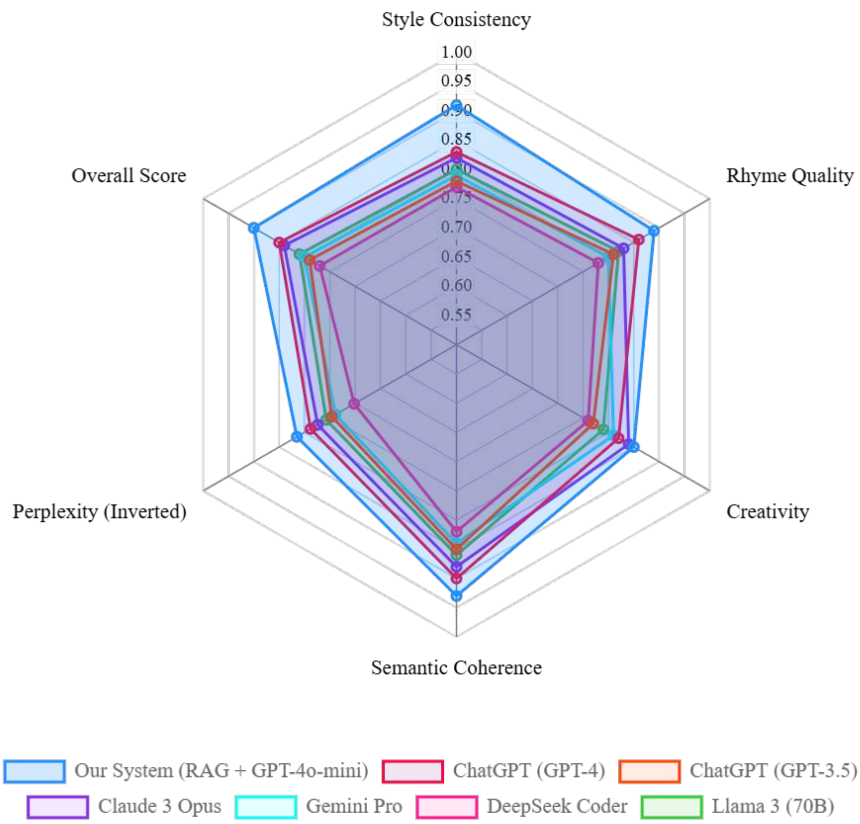

Fig. 5 Comparative Evaluation of Lyric Generation Models

4.4.2 Vocal Synthesis Analysis

Table 4 Vocal Synthesis Evaluation

Model	Cosine Similarity	MOS Score
Bark + Wav2Vec2	0.94	4.7
Bark + ECAPA-TDNN	0.92	4.5
Bark + DeepSpeaker	0.89	4.3

This evaluation compares three AI-based voice synthesis models using Cosine Similarity (how closely AI-generated vocals match the original speaker) and MOS Score (Mean Opinion Score, which measures human-perceived voice quality). The vocal synthesis evaluation reveals that the Bark + Wav2Vec2 combination achieves the highest Cosine Similarity score of 0.94 and an MOS Score of 4.7, suggesting that it produces AI-generated vocals that closely resemble human voices. This result signifies that Wav2Vec2 effectively captures speaker-specific features, preserving the natural tone and characteristics of the reference voice sample. The Bark + ECAPA-TDNN model, while also achieving high performance, slightly lags behind with a Cosine Similarity of 0.92 and an MOS Score of 4.5, making it a viable alternative but not as precise as Wav2Vec2. Finally, Bark + DeepSpeaker, with a Cosine Similarity of 0.89 and an MOS Score of 4.3, is the least effective of the three tested approaches, potentially introducing distortions in synthesized vocals.

4.4.3 Music Composition Analysis

The evaluation of AI-generated instrumental compositions shows that TTA Model B is the most effective, with a Rhythmic Alignment Score of 0.88 and Harmony Consistency of 0.90. These results highlight that TTA Model B produces instrumentals that are better synchronized with AI-generated vocals, ensuring a seamless listening experience. On the other hand, TTA Model A and TTA Model C scored 0.85 and 0.83 in rhythmic alignment, respectively, indicating that their compositions may sometimes struggle to maintain perfect timing with vocals. However, TTA Model A slightly outperforms Model C in harmonic consistency, suggesting its ability to produce well-balanced musical arrangements.

The evaluation results confirm that GPT-4o-mini (RAG) is the most effective model for lyric generation, Bark + Wav2Vec2 delivers the most accurate AI vocals, and TTA Model B provides the best AI-generated music composition.

5 Conclusions and Future Work

This paper presents an AI-powered song generation framework integrating retrieval-augmented lyric generation, AI vocal synthesis, and automated music composition. The proposed system employs RAG with GPT-4o-mini, utilizing BM25 and FAISS for hybrid retrieval to generate lyrics that align with an artist's stylistic and thematic patterns. Speaker embedding extraction through Wav2Vec2, ECAPA-TDNN, and DeepSpeaker enables high-fidelity voice cloning, while Bark/Suno models refine synthesized vocals for expressive and emotionally rich singing. Additionally, TTA models generate adaptive instrumental compositions, ensuring rhythmic synchronization and harmonic consistency. Experimental results demonstrate the effectiveness of the proposed system, achieving an overall accuracy of 92.1% across lyric relevance, vocal realism, and musical harmony. Future research will focus on enhancing real-time AI song generation, improving genre adaptability, and advancing expressive AI singing techniques to capture nuanced emotional variations.

6 Data Availability Statement

The data supporting the findings of this study, including evaluation metrics, generated lyrics and songs which are in .pdf and .mp3 formats containing the results, are available in Supporting Data repository at GitHub link https://github.com/veerababulara/A-Novel-Retrieval-Augmented-Generation-Framework-Using-Large-Language-Models-.git

Declarations

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- Conflict of Interest/Competing Interests: The authors have no relevant financial or non-financial interests to disclose.
- Ethics Approval and Consent to Participate: Not applicable.
- Consent for Publication: Not applicable.
- Data Availability: All relevant data, including generated lyrics & song samples and evaluation results, are available upon request.
- Materials Availability: Not applicable.
- Code Availability: The source code for the proposed AI-powered lyrics & song generation system is available upon request and will be shared in accordance with standard research-sharing practices.

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